SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Initialization and Planning Phase

Date	2 july 2026
Team ID	xxxxxx
Project Title	Smart Lender
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to improve the loan approval process using Artificial Intelligence and Machine Learning. SmartLender predicts loan eligibility by analyzing applicant details such as income, education, employment status, loan amount, credit history, and property area. The system helps banks make faster, more accurate, and data-driven lending decisions while reducing manual effort and processing time.

Project Overview	
Objective	The primary objective is to develop an AI-powered loan eligibility prediction system using machine learning techniques that enables banks and financial institutions to make faster, accurate, and reliable loan approval decisions.
Scope	The project focuses on automating loan eligibility prediction through data preprocessing, machine learning model development, model evaluation, Flask web application integration, and cloud deployment for real-time predictions..
Problem Statement	
Description	Traditional loan approval processes rely on manual verification, making them time-consuming, error-prone, and inconsistent. Evaluating a large number of loan applications efficiently remains a major challenge for financial institutions.
Impact	Implementing an AI-based loan prediction system improves prediction accuracy, reduces processing time, minimizes financial risk, supports better decision-making, and enhances customer satisfaction.
Proposed Solution	
Approach	Develop a machine learning-based loan eligibility prediction system that analyzes applicant information and predicts loan approval. Multiple classification algorithms are trained and evaluated, with the best-performing XGBoost model integrated into a Flask web application for real-time predictions.
Key Features	- AI-powered loan eligibility prediction. - Applicant data analysis using machine learning.

	- Real-time decision-making for quicker loan approvals. - Fast, accurate, and reliable decision support.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Processor	Intel Core i3 or above	Intel Core i3 or above
RAM	System Memory	4 GB (8 GB Recommended)
Storage	Free Disk Space	10 GB Free Space
System Type	Operating System Architecture	64-bit OS
Display	Screen Resolution	1366 × 768 or Highe
Software		
Operating System	Supported Platforms	Windows 10/11, Linux, macOS
Python Environment	Environment Management	Anaconda Navigator
Programming Language	Core Language	Python 3.8+
Framework	Web Framework	Flask
Libraries	ML & Data Science Libraries	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, XGBoost
Code Editor	Development IDE	PyCharm / VS Code
Web Browser	Browser Support	Google Chrome / Microsoft Edge
Data		
Data	Source, size, format	Loan Eligibility Dataset CSV Format